

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

06

FINITE AUTOMATA WITH OUTPUTS

DENNIS GUNAWAN

REVIEW

Minimization of DFA:

Minimization of DFA

Partitioning Method

Table Filling Method (Myhill Nerode Theorem)

OUTLINE

Mealy Machine

Moore Machine

Conversion of Moore Machine to Mealy Machine

Conversion of Mealy Machine to Moore Machine



MEALY MACHINE

- Six-tuple notation: $M = (Q, \Sigma, \delta, q_0, \Delta, \lambda)$
 - Q : a set of states
 - Σ : a set of input symbols
 - δ : a transition function: $Q \times \Sigma \rightarrow Q$
 - q_0 : a start state
 - Δ : a set of outputs
 - λ : an output function: $\Sigma \times Q \rightarrow \Delta$

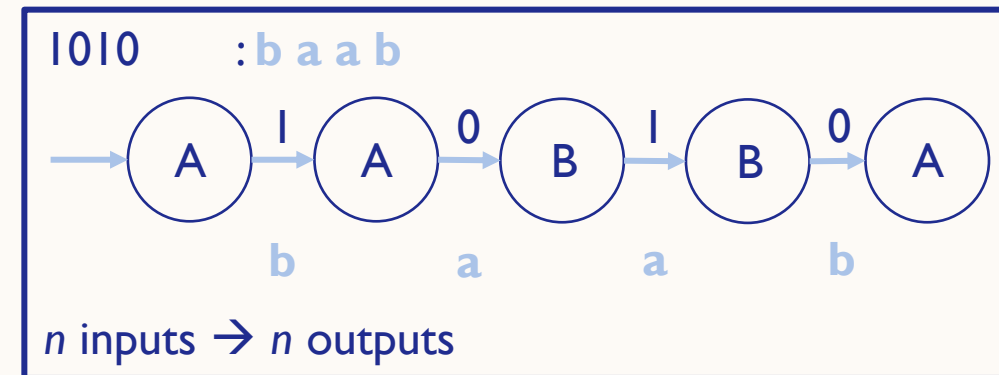
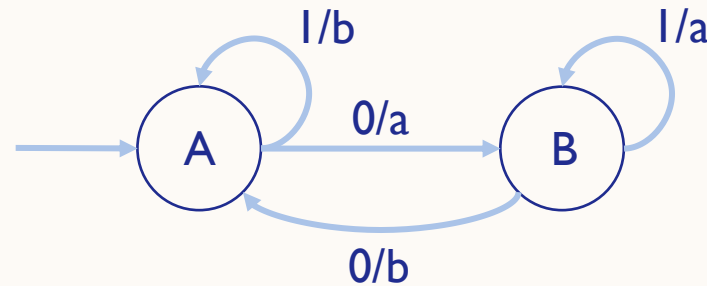
$$\Delta = \{a, b\}$$

$$\lambda(A, 0) = a$$

$$\lambda(A, 1) = b$$

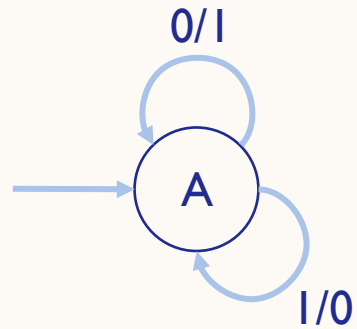
$$\lambda(B, 0) = b$$

$$\lambda(B, 1) = a$$



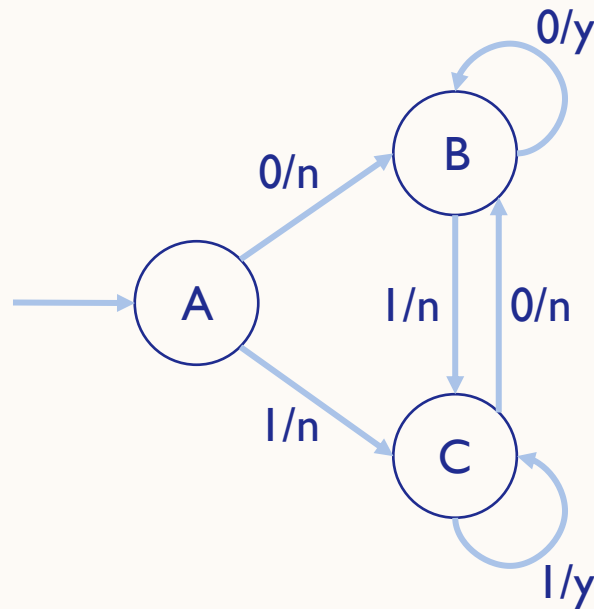
MEALY MACHINE

Construct a Mealy Machine that produces the 1's Complement of any binary input string.



MEALY MACHINE

Design a Mealy Machine accepting the language consisting of strings from Σ^* , where $\Sigma = \{0,1\}$ and the strings should end with either '00' or '11' (print 'y'; otherwise, print 'n').



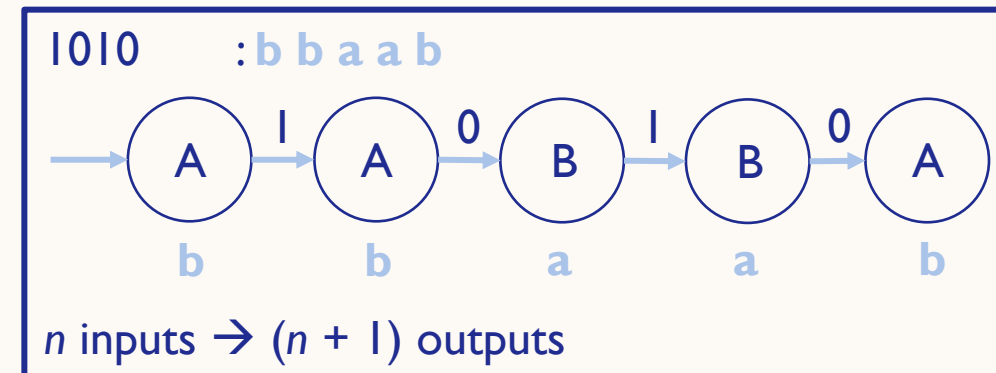
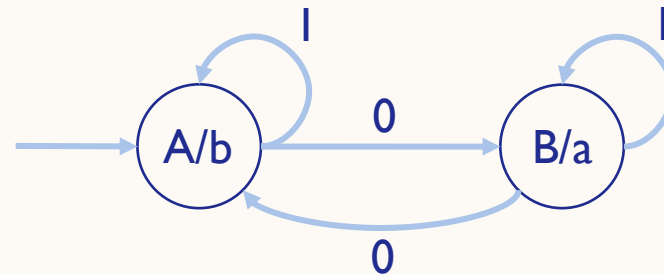
MOORE MACHINE

- Six-tuple notation: $M = (Q, \Sigma, \delta, q_0, \Delta, \lambda)$
 - Q : a set of states
 - Σ : a set of input symbols
 - δ : a transition function: $Q \times \Sigma \rightarrow Q$
 - q_0 : a start state
 - Δ : a set of outputs
 - λ : an output function: $Q \rightarrow \Delta$

$$\Delta = \{a, b\}$$

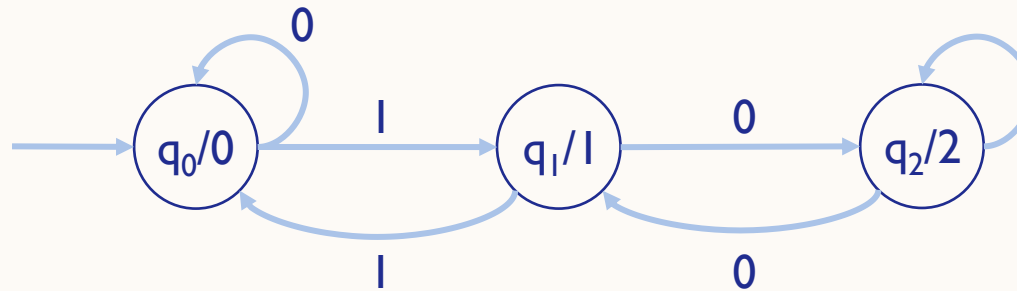
$$\lambda(A) = b$$

$$\lambda(B) = a$$



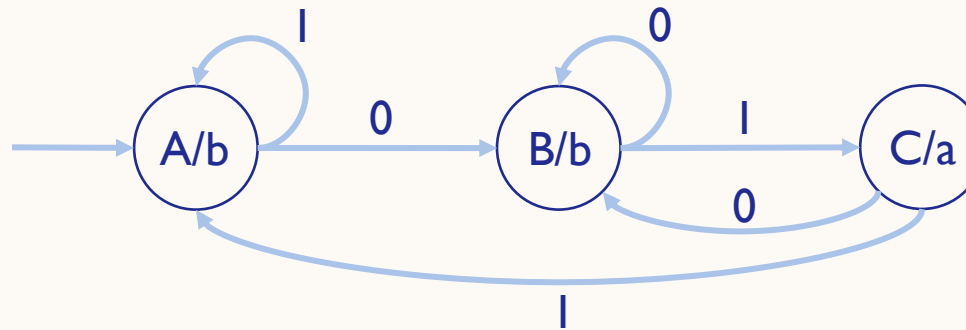
MOORE MACHINE

Design a Moore Machine that prints the residue modulo 3 of any binary input string interpreted as a binary integer.



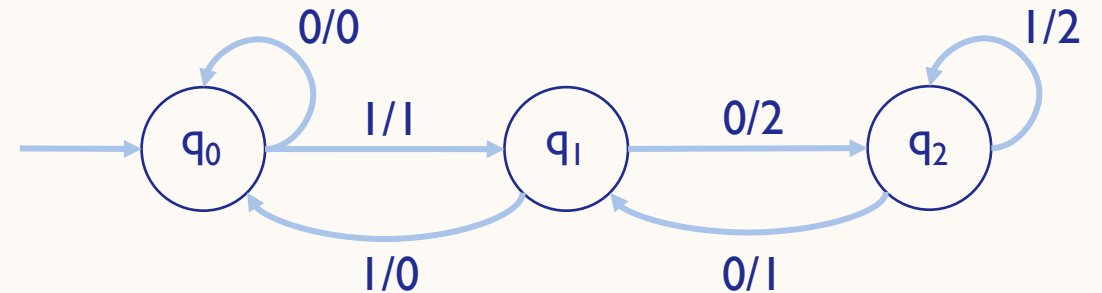
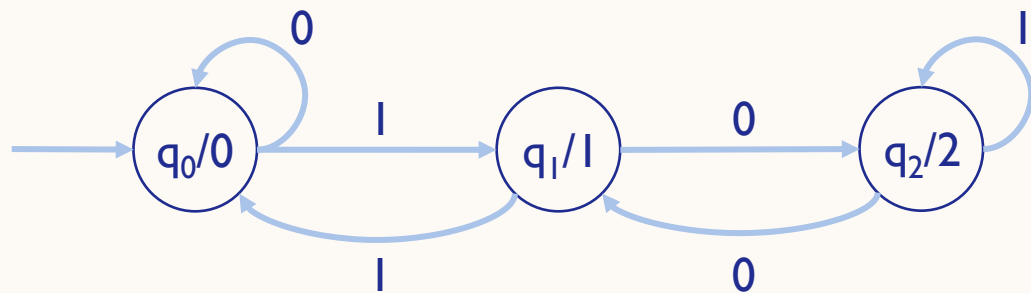
MOORE MACHINE

Construct a Moore Machine that prints 'a' whenever the sequence '01' is encountered in any binary input string.



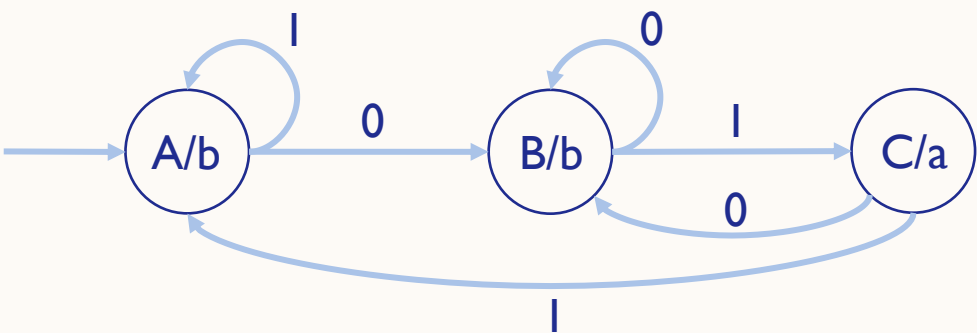
CONVERSION OF MOORE MACHINE TO MEALY MACHINE

Convert the Moore Machine below to its equivalent Mealy Machine.

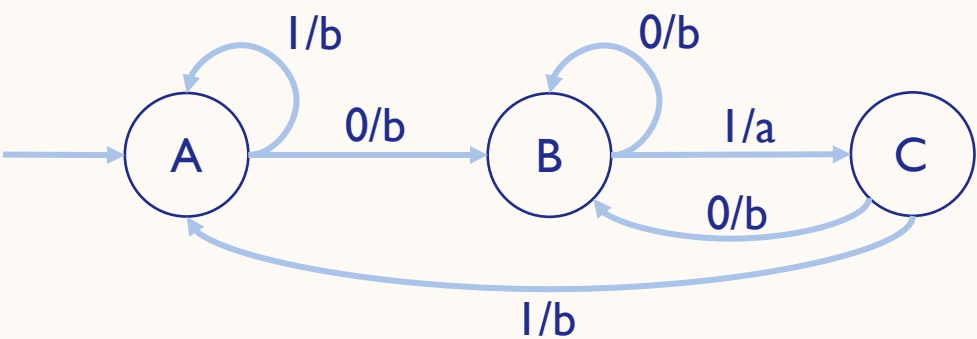


CONVERSION OF MOORE MACHINE TO MEALY MACHINE

Convert the Moore Machine below to its equivalent Mealy Machine.



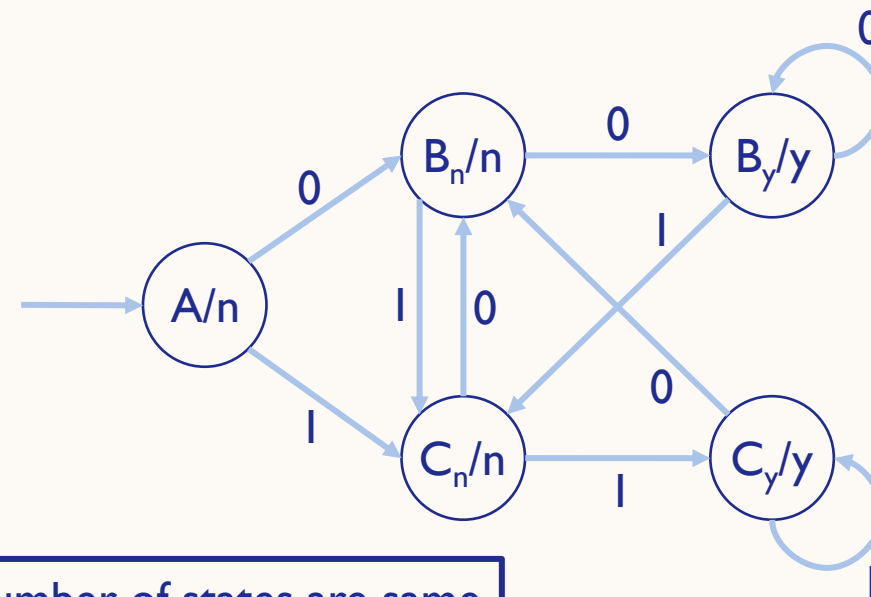
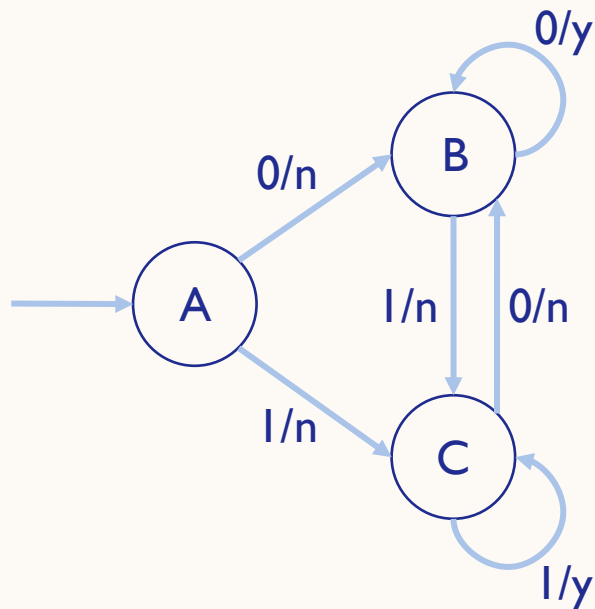
State	0	1	Output
→A	B	A	b
B	B	C	b
C	B	A	a



State	0	1
→A	B, b	A, b
B	B, b	C, a
C	B, b	A, b

CONVERSION OF MEALY MACHINE TO MOORE MACHINE

Convert the Mealy Machine below to its equivalent Moore Machine.



- Moore → Mealy : number of states are same
- Mealy → Moore: number of states increase
 - Mealy : x states and y outputs
 - Moore : max (x * y) number of states



PRACTICE

EXERCISES

1. Construct a Mealy Machine that gives 2's Complement of any binary input string. Assume that the last carry bit is neglected.
Input: LSB $\rightarrow$ MSB
2. Design a Mealy Machine that gives the decimal value of any binary input string interpreted as a binary integer. Then convert it to its equivalent Moore Machine.
 $\Sigma = \{0,1\}$
 $\Delta = \{0,1,2,3,4,5,6,7\}$
3. Design a Moore Machine that prints the residue modulo 5 of any binary input string interpreted as a binary integer.

EXERCISES

4. For the following Moore Machine, the input alphabet is $\Sigma = \{a, b\}$ and the output alphabet is $\Delta = \{0, 1\}$. Run the following input sequences and find the respective outputs.
- i. aabab
 - ii. abbb
 - iii. ababb

State	a	b	Output
$\rightarrow q_0$	q_1	q_2	0
q_1	q_2	q_3	0
q_2	q_3	q_4	1
q_3	q_4	q_4	0
q_4	q_0	q_0	0

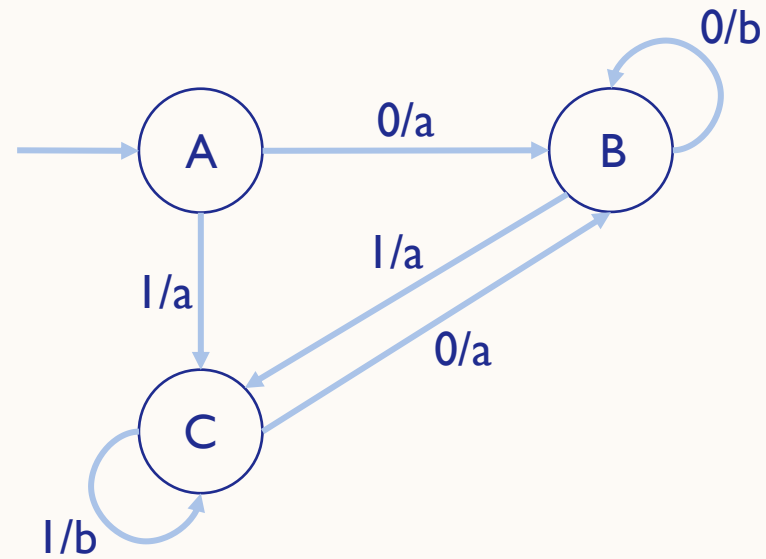
EXERCISES

5. Construct a Moore Machine that counts the occurrences of the sequence 'abb' in any input strings over $\{a, b\}$ (print '1'; otherwise, print '0'). Then convert it to its equivalent Mealy Machine.
6. Convert the given Moore Machine to its equivalent Mealy Machine.

State	0	1	Output
$\rightarrow q_0$	q_1	q_2	1
q_1	q_3	q_2	0
q_2	q_2	q_1	1
q_3	q_0	q_3	1

EXERCISES

7. Convert the following Mealy Machine to its equivalent Moore Machine.



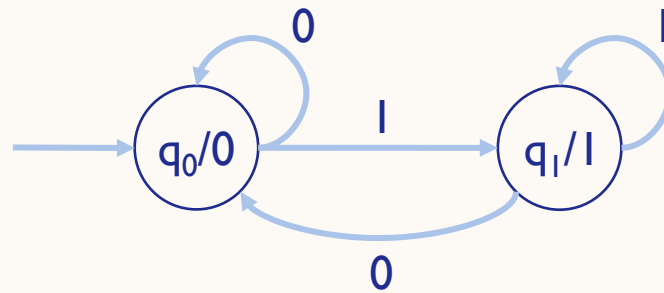
EXERCISES

8. Convert the given Mealy Machine to its equivalent Moore Machine.

State	a	b
$\rightarrow q_0$	$q_3, 0$	$q_1, 1$
q_1	$q_0, 1$	$q_3, 0$
q_2	$q_2, 1$	$q_2, 0$
q_3	$q_1, 0$	$q_0, 1$

EXERCISES

9. Describe what the following Moore Machine is for.



REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

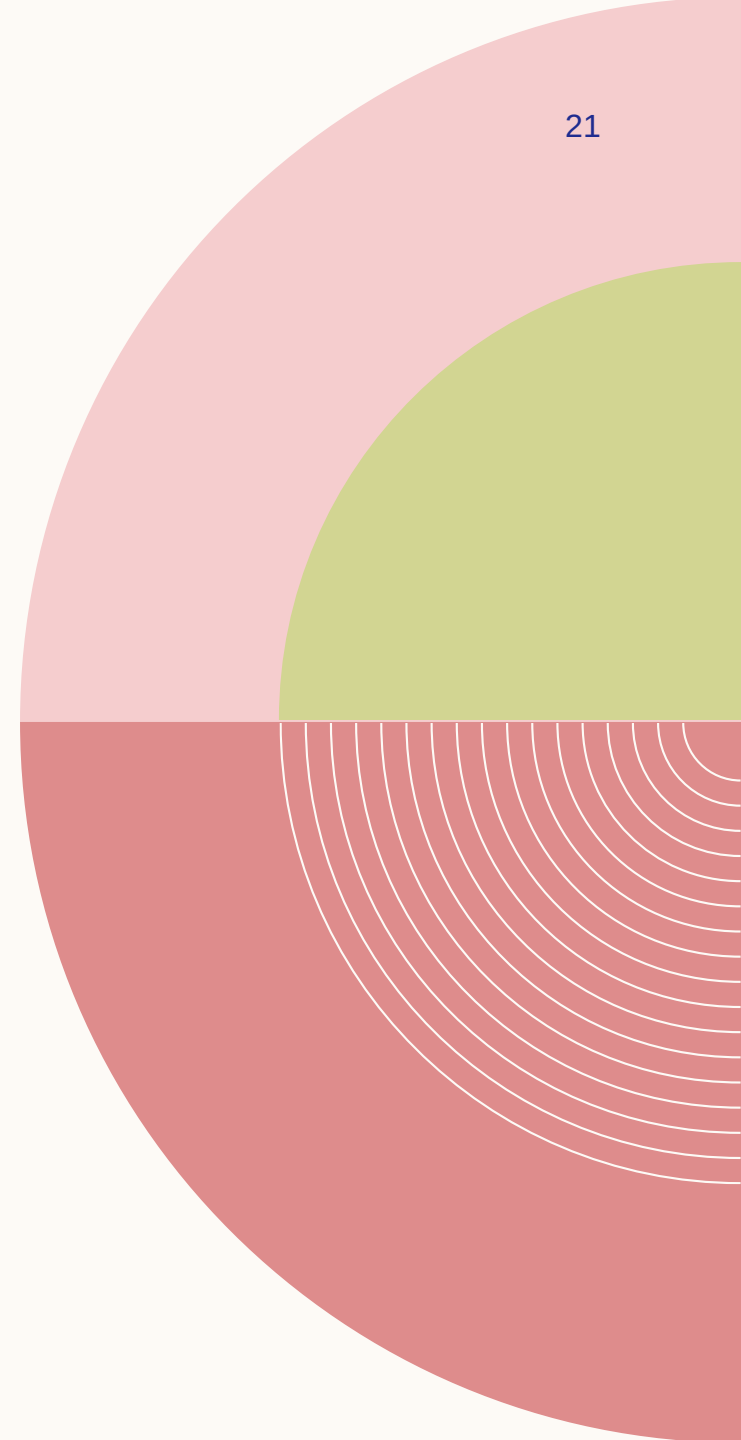
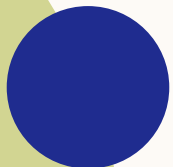
Regular Expression:

Identities of Regular Expression

Arden's Theorem

Conversion of FSA to Regular Expression

Conversion of Regular Expression to FSA



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



iF INFORMATIKA
UMN